

Infrared Spectroscopy and Mass Spectrometry

Answer key with worked solutions

A. IR spectroscopy

1. IR basics.

(a) The bond **vibrates** more strongly (stretches or bends): it absorbs the energy of IR radiation at a wavenumber characteristic of that bond. (b) The region **below about 1500 cm⁻¹**: a complex pattern unique to each compound, **compared with a database** of known spectra to confirm an identity.

2. Functional groups.

(a) **alcohol** (O-H and C-O) (b) **C=O** with no O-H: a ketone, aldehyde or ester (c) **carboxylic acid** (very broad O-H and C=O) (d) **nitrile**, C≡N

3. Wavelengths.

$$\begin{aligned}\text{(a) } \lambda &= \frac{1}{3300 \text{ cm}^{-1}} = 3.030 \times 10^{-4} \text{ cm} \times \frac{10^4 \mu\text{m}}{1 \text{ cm}} = \mathbf{3.03 \mu\text{m}} \\ \text{(b) } \lambda &= \frac{1}{2250 \text{ cm}^{-1}} = 4.444 \times 10^{-4} \text{ cm} = \mathbf{4.44 \mu\text{m}}\end{aligned}$$

4. Three C₃ compounds.

Propan-1-ol: broad O-H at 3200–3550 cm⁻¹, **no C=O**. **Propanal**: strong C=O at about 1730 cm⁻¹, **no O-H**. **Propanoic acid**: **both** a very broad O-H (2500–3300 cm⁻¹) and a C=O.

B. Mass spectrometry

5. Butanone.

72: molecular ion [CH₃COCH₂CH₃]⁺ (C₄H₈O, 72). **57**: CH₃CH₂CO⁺ (loss of CH₃, 72 – 15). **43**: CH₃CO⁺ (loss of C₂H₅, 72 – 29). **29**: C₂H₅⁺.

6. Halogens.

3 : 1 for M and M+2: **chlorine** (one Cl atom). 1 : 1: **bromine** (one Br atom).

7. Bromine.

$$A_r = 78.918 \times 0.5069 + 80.916 \times 0.4931$$
$$= \mathbf{79.90}$$

C. Challenge

8. The unknown.

Ethanoic acid, CH_3COOH . Very broad $2500\text{--}3300\text{ cm}^{-1}$: **acid O-H**; 1710 cm^{-1} : **C=O**. MS: **60** = molecular ion ($\text{C}_2\text{H}_4\text{O}_2$, 60 g/mol); **45** = COOH^+ (loss of CH_3); **43** = CH_3CO^+ (loss of OH , $60 - 17$); **15** = CH_3^+ .